Name: Nayan Kumar
Roll no: 2401010228
Subject: DBMS
Course: B.Tech CSE

Section - A

Ques1 - What is database management system? Why is it required?

Ans - A database management system is a software that allows users to create, store, manage, retrieve and manipulate data in a structured and efficient way.

DBMS is required because:

a) Reduces data redundancy
b) Avoids data inconsistency
c) Provides data security
d) Data integrity
e) Easy data retrieval

Ques2 - Why is DBMS better than the traditional file system?

Ans - DBMS is better than the traditional file system because

a) Reduces data redundancy
b) Avoids data inconsistency
c) Better data security
d) Backup & recovery
e) Data sharing and concurrency control.

Ques3 - What is data abstraction? Explain its levels.

Ans - Data abstraction in DBMS is the process of hiding complex internal details of how data is stored and maintained and showing only the necessary information to users.

Levels of data abstraction:

1. Physical Level (Internal Level)

- the lowest level of abstraction.
- describes how data is actually stored in memory.
- includes details like storage structures, indexing, file organization, etc.

2. Logical Level (Conceptual Level)
- describes what data is stored in the database.
- defines relationships among data.
- does not show physical storage details.

3. View Level (External Level)
- the highest level of abstraction
- shows only required data to specific users.
- different users can have different views of the same database.

Ques 4 - What is the 3 schema architecture? Why is it important?
Ans - The three schema architecture is the database design framewor that separates the database system into 3 different levels to achie data abstraction and independence.

The three schema are:

1. External Schema (View Level)
- Describes how individual users view the data.
- Different users can have different views
- Hides the rest of the database.

2. Conceptual Schema (Logical Level)
- Describes the overall structure of the database.
- Includes tables, attributes, relationships and constraints.
- Does not include physical storage details.

3. Internal Schema (Physical Level)
- Describes how data is physically stored in memory.
- Including indexing, storage structure, file organization, etc.

Why is it important:
- Provides data abstraction
- Ensures data independence
- Improves security.

What is data independence? Why is it needed?

Data independence is the ability to modify the database schema at one level without affecting the schema at the next higher level.

Why is it needed:

- Easier database maintenance
- Flexibility
- Reduced development cost
- System stability

Ques6 - What is an Entity-Relationship (ER) model?

Ans - The Entity Relationship model is a high level conceptual data model used to design and represent the structure of a database visually. It was introduced by Peter Chen in 1976.

The ER model describes data using:

- Entities
- Attributes
- Relationships

It is mainly used during the database design phase.

Ques7 - What are Integrity constraints? Why are they important in DBMS?

Ans - Integrity constraints are rules in a DBMS that ensure the accuracy, validity and consistency of data stored in a database. They prevent invalid data from being entered into tables.

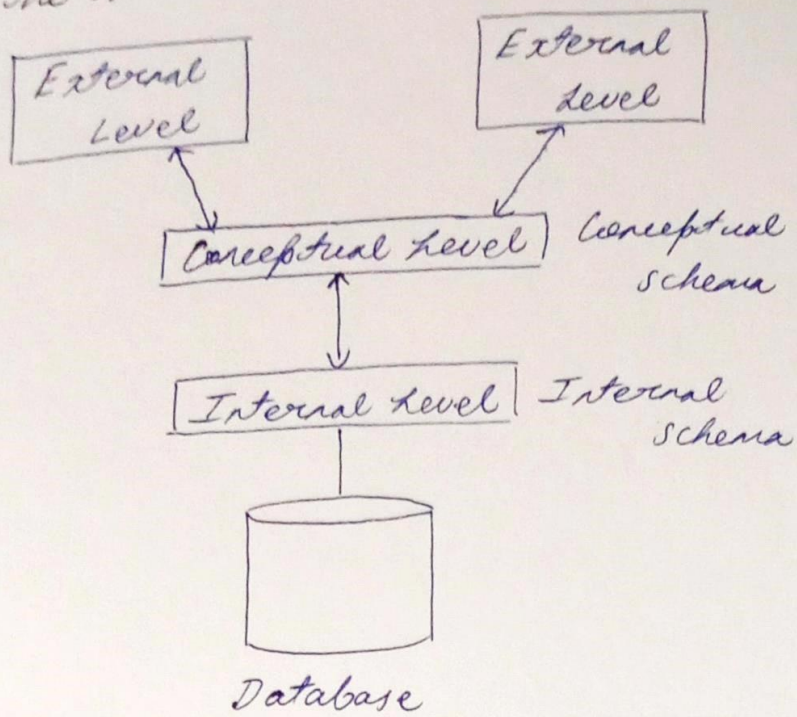
Why integrity constraints are important:

- Maintain data accuracy.
- Ensure consistency
- Improve reliability
- Prevents data corruption.

Section-B

Ques1 - Draw the three schema architecture and explain each

Ans -



a) External Schema (New Level)

- Describes how individual users view data.
- Different users can have different views.
- Hides the rest of the database.

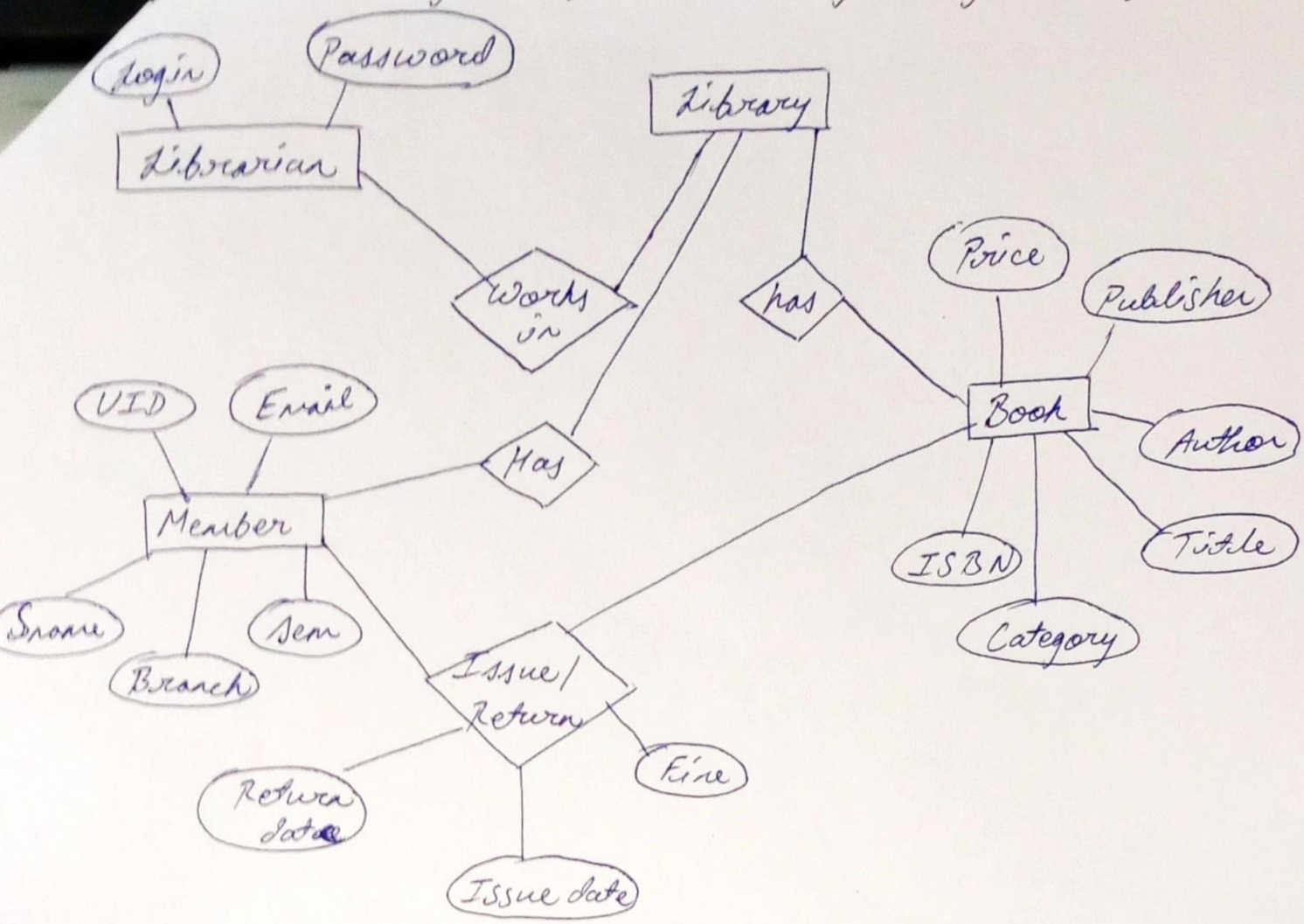
b) Conceptual Schema (Logical Level)

- Describes the overall structure of the database.
- Includes tables, attributes, relationships, constraints
- Does not include physical storage details.

c) Internal Schema (Physical Level)

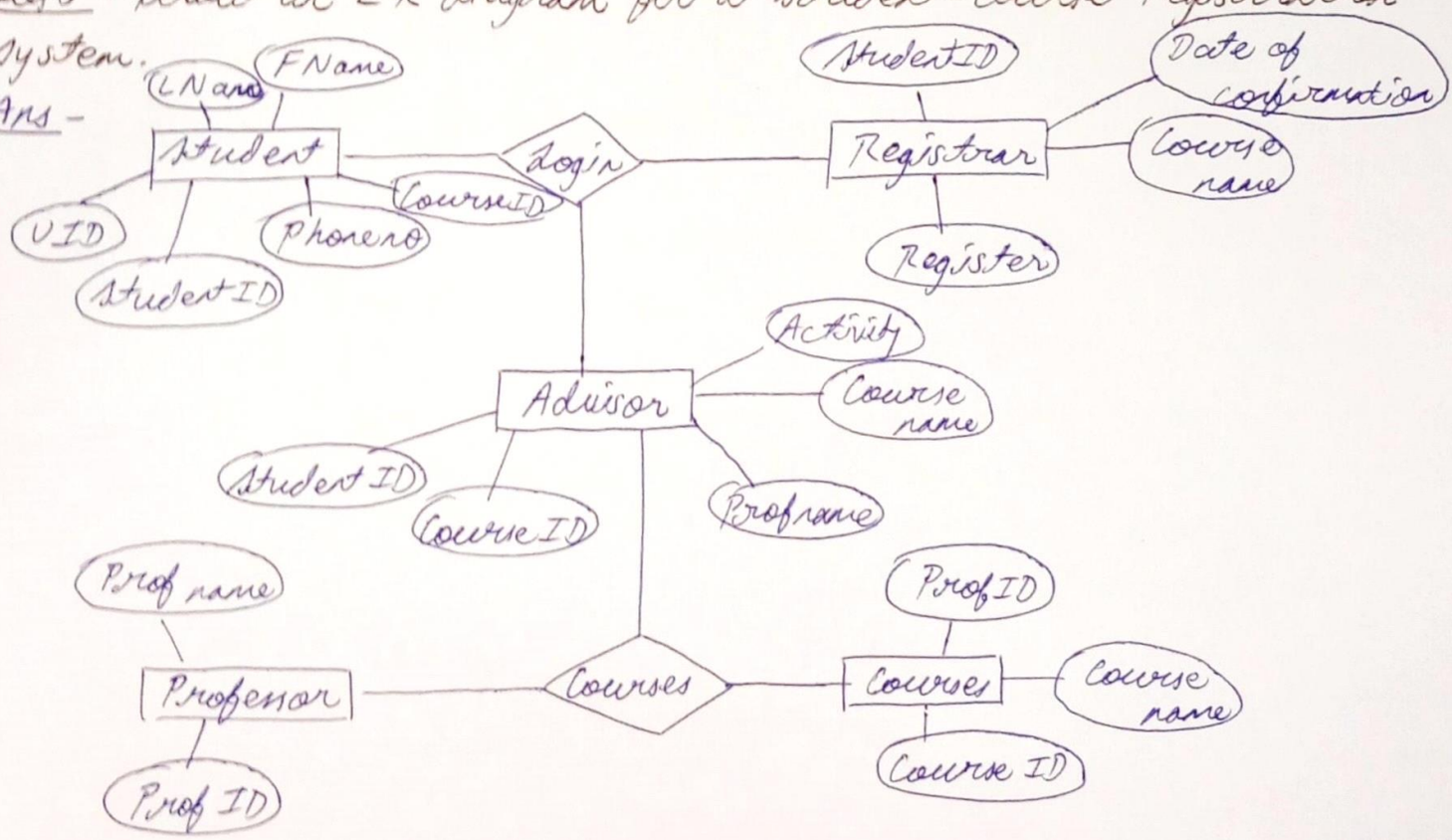
- Describes how data is physically stored in memory.
- Includes indexing, storage structure, file organization, etc.

...w an ER diagram for a library management system.



Ques 3 - Draw an ER diagram for a Student-Course Registration system.

Ans -



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Section-C

Ques1 - A hospital wants to store patient, doctor, appointment. Identify entities, attributes and relationships.

Ans - Entities

1. Patient
2. Doctor
3. Appointment

Attributes

1. Patient_ID
2. Name
3. Age
4. Gender
5. Address
6. Phone number
7. Doctor_ID
8. Name
9. Specialization
10. Phone_number
11. Experience
12. Appointment_ID
13. Appointment_Date
14. Appointment_Time
15. Status

Relationships

1. Patient — books — Appointment
- A patient can book many appointments
- Each appointment belongs to one patient
- one to many relationship.

2. Doctor — attends — Appointment
- A doctor can attend many appointments.
- Each appointment is handled by one doctor.
- one to many relationship.

Ques2 - An online shopping platform stores customer, product and order data. Suggest keys and constraints.

Ans - 1. Customer Entity: Customer (Customer_ID, Name, Email, Phone, Address)

Keys:
- Primary Key: Customer_ID
- Candidate Key: Email should be unique

Constraints
- Customer_ID → NOT NULL, UNIQUE
- Name → NOT NULL
- Email → UNIQUE, NOT NULL
- Phone → UNIQUE (optional but recomanended)

Entity: Product (Product_ID, Product_Name, Price, Stock, category)

Primary Key: Product_ID

Constraints:

- Product_ID → NOT NULL, UNIQUE
- Price → CHECK (Price > 0)
- Stock → CHECK (Stock >= 0)
- Product_Name → NOT NULL

3. Order Entity: Order (Order_ID, Order_Date, Customer_ID, Total_Amount, Status)

Keys:

- Primary Key: Order_ID
- Foreign Key: Customer_ID

Constraints:

- Order_ID → NOT NULL, UNIQUE
- Customer_ID → FOREIGN KEY
- Total_Amount → CHECK
- Order_Date → NOT NULL
- Status → CHECK

Ques3- A university stores students records. Explain how data abstraction helps different users.

Ans - 1. View Level (External level) → At this level, each user sees only the data relevant to them.

Teacher:

- can see: Student_ID, Name, course, marks, Attendance
- cannot see: Fee details

Accounts Dept.

- can see: Student_ID, Name, Fee_status
- cannot see: marks

Student:

- can see: own marks, attendance, fee status
- cannot see: other student's records

Benefits: Ensures security and Privacy

2. Logical Level (Conceptual Level):
- Defines the complete structure of the database
- Includes tables like: student (Student_ID, Name, Courses, marks, Attendance)
- Managed by database designers and administrators

Benefits: Maintains overall organization and relationships.

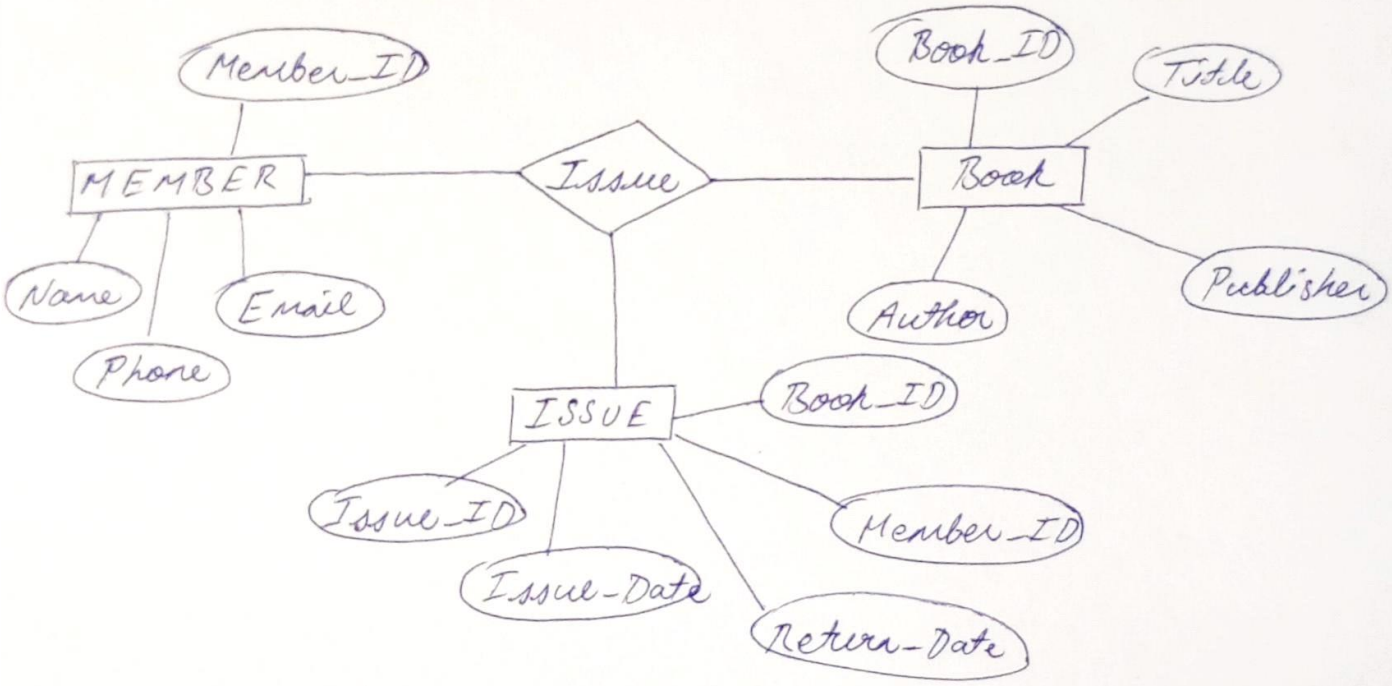
3. Physical Level (Internal Level):
- Describes how data is stored in memory.
- Includes indexing, file organization, storage, blocks, etc.
- Handled by DBMS internally

Benefits: Users do not need to know how data is physically stored.

Section-D

Ques1 - Draw a simple ER diagram for any real life system and explain one key design decision.

Ans:



Entities & Attributes:

1. Member
- Member-ID
- Name
- Phone
- Email

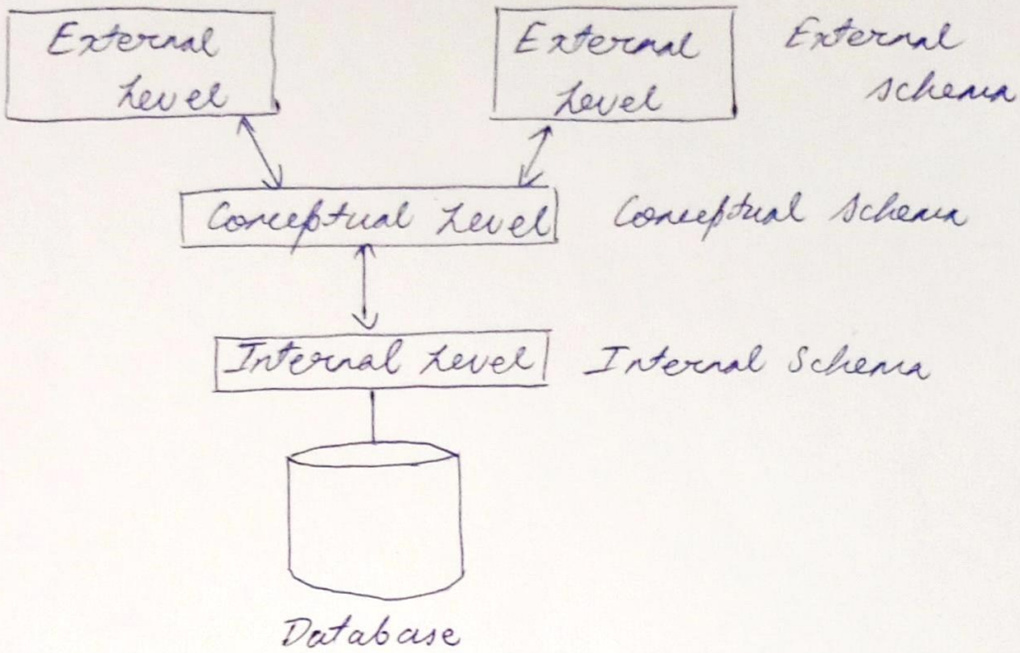
2. Book
- Book-ID
- Title
- Author
- Publisher

3. Issue
- Issue-ID
- Issue-Date
- Return-Date
- Member_ID
- Book_ID

can issue many books.
can be issued by many members (at different times).
creates a many-to-many relationship b/w member and book.
resolve this, we introduce the ISSUE Entity.

Ques2 - Draw the three schema ~~artitec~~ architecture and mention one benefit.

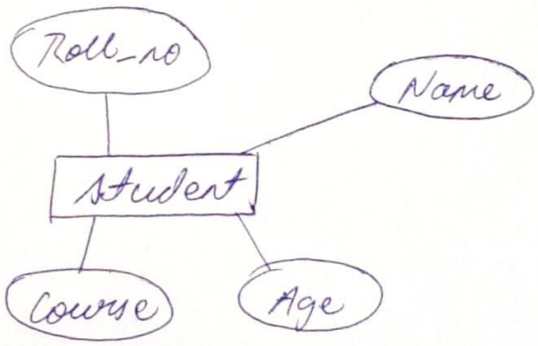
Ans -



Benefit: Data-Independence:- changes made at the physical level (like changing storage method) do not affect the logical or external levels).

Ques3 - Draw an entity with attributes and identify its Primary key.

Ans -



Entity: Student (Represented by a rectangle)

Attributes:
- Roll_no
- Name
- Course
- Age

Primary Key: Roll_no
- Uniquely identifies each student
- cannot be NULL.
- No two students can have same roll no.